

Ritesh Gharat

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Education

Vidyavardhini's College of Engineering and Technology, Mumbai

B.E. in Computer Engineering

CGPA: 8.34

2023 – 2027

Experience

Software Engineering Intern, Axisol

Jul 2025 – Oct 2025

- Built and optimized backend systems using Node.js and TypeScript with focus on performance and scalability.
- Improved API response latency and system efficiency through query optimization and caching strategies.
- Collaborated on production-grade distributed systems and debugging of real-time issues.

Projects

TAMX – ESP32-Based Handheld Multi-Tool System

- Built a handheld embedded system using ESP32 with SSD1306 OLED (SPI) and 6-button hardware interface.
- Designed a modular menu-driven UI with state machines for efficient navigation and low memory usage.
- Implemented hardware input handling using GPIO with debouncing and interrupt-driven design.
- Developed Wi-Fi utilities and HTTP-based communication for real-time device interaction.
- Built a PPT remote system using REST endpoints (`/next`, `/prev`, `/start`) with Python server backend.
- Integrated system-level features including local AI API support and multi-mode operation.

PayProtect – On-Device Scam Detection System

- Designed a real-time, on-device inference system using TensorFlow Lite for resource-constrained environments.
- Optimized CNN model achieving 97.1% accuracy with inference latency <20ms.
- Integrated Android Accessibility Service for low-level system interaction and monitoring.
- Focused on memory-efficient deployment and edge AI optimization.

Sleep Apnea Detection using ECG Signals

- Developed ML models for biomedical signal processing using ECG datasets.
- Implemented signal preprocessing, feature extraction, and classification pipelines.
- Deployed models via FastAPI with emphasis on performance and scalability.

Hackathons

AMD AI Sprint (IIT Bombay)

- Fine-tuned LLaMA-4B model and optimized training on AMD MI300X GPUs.
- Focused on compute optimization and efficient resource utilization.

Achievements

- First Prize, Techblitz Hackathon (VCET)
- First Prize, Oscillation Research Presentation 2025
- Published Research Paper (INCET 2025) on Sleep Apnea Detection

Skills

Languages: C++, Python, C (Basics), SQL, Javascript, TypeScript

Embedded/Systems: ESP32, Arduino, GPIO, Interrupts, State Machines, Edge AI

Machine Learning: TensorFlow, TFLite, Signal Processing

Tools: Linux, Git, Docker, KiCad

Concepts: Data Structures, Operating Systems, Computer Networks, Memory Optimization, Low-Latency Systems